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Assessing the relationship between cyanobacterial blooms and respiratory-related hospital visits: Green Bay, Wisconsin 2017–2019

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Abstract

Potential acute and chronic human health effects associated with exposure to cyanobacteria and cyanotoxins, including respiratory symptoms, are an understudied public health concern. We examined the relationship between estimated cyanobacteria biomass and the frequency of respiratory-related hospital visits for residents living near Green Bay, Lake Michigan, Wisconsin during 2017–2019. Remote sensing data from the Cyanobacteria Assessment Network was used to approximate cyanobacteria exposure through creation of a metric for cyanobacteria chlorophyll-a (Chl_{BS}). We obtained counts of hospital visits for asthma, wheezing, and allergic rhinitis from the Wisconsin Hospital Association for ZIP codes within a 3-mile radius of Green Bay. We analyzed weekly counts of hospital visits versus cyanobacteria,, which was modelled as a continuous measure (Chl_{BS}) or categorized according to World Health Organization's (WHO) alert levels using Poisson generalized linear models. Our data included 2,743 individual hospital visits and 114 weeks of satellite derived cyanobacteria biomass indicator data. Peak values of Chl_{BS} were observed between the months of June and October. Using the WHO alert levels, 60% of weeks were categorized as no risk, 19% as Vigilance Level, 15% as Alert Level 1, and 6% as Alert Level 2. In Poisson regression models adjusted for temperature, dewpoint, season, and year, there was no association between Chl_{BS} and hospital visits (rate ratio [RR] [95% Confidence Interval (CI)] = 0.98 [0.77, 1.24]). There was also no consistent association between WHO alert level and hospital visits when adjusting for covariates (Vigilance Level: RR [95% CI] 0.88 [0.74, 1.05], Alert Level 1: 0.82 [0.67, 0.99], Alert Level 2: 0.98 [0.77, 1.24], compared to the reference no risk category). Our methodology and model provide a template for future studies that assess the association between cyanobacterial blooms and respiratory health.

Keywords: Cyanobacteria, Harmful algal blooms, CyanoHABs, Satellite imagery, Respiratory health, Electronic Health Records (EHRs).

1. Introduction

Cyanobacterial harmful algal blooms (CyanoHABs) are changing in magnitude (Mishra et al., 2019), frequency (Coffer et al., 2021), and spatial extent (Schaeffer et al., 2022) and may put humans at a greater risk for exposure and associated illnesses (Davis & Gobler, 2016; Gilbert, 2017). Few epidemiologic studies have examined the significance and magnitude of health effects associated with exposure to cyanoHABs and associated cyanotoxins.

Previous work on *Karenia brevis* blooms along the coast of Florida, U.S., identified human health effects from aerosolized brevetoxins in marine and coastal algal blooms, including respiratory irritation and asthma exacerbations (Backer et al., 2005; Bean et al., 2011; Kirkpatrick et al., 2011). However, it is unclear whether aerosolized freshwater cyanobacteria or their associated toxins can produce similar effects. National and state-based public health surveillance systems have documented respiratory health complaints related to nose, throat, and ear irritation following a harmful algal bloom (HAB) event (Backer et al., 2015; Centers for Disease Control and Prevention (CDC), 2022). Additionally, studies have shown that cyanotoxins have been detected in the air and in the human upper respiratory system following exposure (Backer et al., 2010; Hu et al., 2020; May et al., 2018; Olson et al., 2020; Schaefer et al., 2020). Microcystin, a cyanotoxin, was detected on nasal swab specimens of individuals who were directly (contacts with the water) and indirectly (being near the water) exposed to a cyanoHAB bloom (Backer et al., 2010; Schaefer et al., 2020). These data indicate that inhaling cyanotoxins is a potential exposure route that warrants further investigation.

Epidemiologic studies assessing the relationship between exposure to cyanobacterial blooms and associated respiratory diseases can be costly, time-consuming, and reliant on unpredictable naturally occurring cyanobacteria bloom events. The purpose of this analysis is to explore the use of existing data

sources to identify patterns in cyanobacteria bloom occurrence and respiratory-related hospital visits over time. We present a methodology for analyzing these relationships using satellite remote sensing data from the Cyanobacteria Assessment Network (CyAN) (Schaeffer et al., 2015) and health data from electronic health records (EHRs).

2. Materials and methods

2.1 Study area

Green Bay is an inlet located in the western part of Lake Michigan, Wisconsin and is one of the largest freshwater estuaries in the United States. The study area included 10 zip codes within a 3-mile radius of the lower region of the bay (Figure 1). The study area was restricted to the southern bay because available satellite data indicated that cyanobacterial biomass tended to be more concentrated in this region. We used a 3-mile radius based on data showing that brevetoxins from *Karenia brevis* can be blown that far inland during bloom events (Kirkpatrick et al., 2010).

2.2 Respiratory health data

The primary health outcomes of interest for this study were asthma, wheezing, and allergic rhinitis. Patient-level hospital discharge data for patients living in the 10 zip codes in the study area were obtained from the Wisconsin Hospital Association Information Center (WHAIC). The WHAIC received EHR data from approximately 153 hospital systems in Wisconsin during 2017–2019. These data are provided to the Wisconsin Department of Health Services (DHS) upon request and justification for use.

We queried the WHAIC database for inpatient, outpatient, emergency department, and ambulatory surgery hospital visits containing a primary International Classification of Diseases, Tenth Revision, Clinical Modification (ICD-10-CM) diagnosis code for asthma (J45.X), wheezing (R06.2), or allergic rhinitis

(J30–J31). Counts of hospital visits for these respiratory ICD-10 codes (hereafter referred to as hospital visits) were aggregated by week for the time period 2017–2019.

2.3 CyAN Satellite data

CyanoHABs Sentinel-3 Ocean and Land Colour Instrument (OLCI) data were first obtained by NASA from the European Space Agency (ESA) through a data sharing agreement. OLCI data were processed by the NASA Ocean Biology Processing Group (OBPG) using their standard satellite ocean color software package (SeaWiFS Data Analysis System, SeaDAS) and a Shuttle Radar Topography Mission (SRTM)-derived 60-m land mask with updates (Schaefer et al., 2020; Urquhart & Schaeffer, 2020). Flags to indicate potential data issues due to cloud, cloud shadow, and adjacency effects from neighboring land pixels, and to identify snow- or ice-covered waterbodies, were also applied (Wynne et al., 2018) and considered as missing data.

The cyanobacteria index (CI_{cyano}) was calculated with a spectral shape algorithm developed by Wynne et al. (2008) and later updated in Lunetta et al. (2015). Cyanobacteria possess both chlorophyll a and phycocyanin where phycocyanin is unique to cyanobacteria in freshwater (Sheath & Wehr, 2015). CI_{cyano} is sensitive to chlorophyll fluorescence and scattering from phycobilin, pigments associated with freshwater cyanobacteria (Wynne et al., 2018), that can be used to estimate concentration (Kutser et al., 2006). A detailed description of the algorithm evolution is described in Coffer et al. (2020). Briefly, the CI_{cyano} uses a spectral shape approach first centered at 681 nm and incorporating neighboring wavelengths at 665 and 709 nm to assess cyanobacterial bloom biomass, and then centered at 665 with neighboring sensor bands at 620 and 681 nm as exclusion criteria to prevent the quantification of non-cyanobacterial blooms. Phycocyanin laboratory methods are not common or standardized at this time for verification of algorithm performance. Therefore, extensive field validation of the CI_{cyano} algorithm

was previously demonstrated across the United States using health advisories (Schaeffer et al., 2018), ecological patterns (Coffer et al., 2020), and agreement with the following: state reported cell and cyanotoxin data (Mishra et al., 2021), state reported events (Whitman et al., 2022), chlorophyll-a (Seegers et al., 2021), and observations from the Unregulated Contaminant Monitoring Rule (Coffer et al., 2021). In addition, the CI_{cyano} has been used effectively in the Caspian Sea, Lake Balaton, and Lakes Taihu and Chaohu in China (Jin et al., 2017; Moradi, 2014; Palmer et al., 2015).

The maximum CI_{cyano} value for each pixel occurring during the 7-day period spanning the 2017–2019 time period was calculated. For each week, two aggregate values were determined within the lake: (1) the total number of pixels in the study area detecting cyanobacterial biomass, and (2) an aggregated weekly median CI_{cyano} value within the study area. The World Health Organization (WHO) (Chorus & Welker, 2021) updated thresholds for cyanobacterial recreational exposure were based on chlorophyll values in cyanobacterial dominated waters; therefore, CI_{cyano} values were converted to chlorophyll (Chl_{BS} in mg m^{-3}) for this analysis using the following equation (Eq. 1) adapted from Seegers et al. (2021).

$$Chl_{BS}\ (mg\ m^{-3}) = 6620 \times CI_{cyano} \qquad \text{Eq. 1}$$

The above equation has the intercept set to zero, whereas the original Chl_{BS} equation included a negative intercept (-3.1±5.2) with a standard deviation range that included zero (Seegers et al., 2021). Because the original intercept range included zero, this value was selected as the intercept to avoid negative Chl_{BS} and to simplify the analysis. Temporal evaluation of pixel detection count and median Chl_{BS} showed that some median Chl_{BS} measurements were skewed when only a small number of pixels were detecting cyanobacterial biomass within the study area. A weighted value was created based on the total percentage of detection pixels versus pixels with non-detection or quality flags to adjust for

skewed pixel detection counts. For each week, the total number of detection pixels were divided by the total number of pixels in the study area. The median Chl_{BS} value for the week was then multiplied by this percentage to obtain a final weighted value. Any week that was missing data for ≥40% of the study area pixels was excluded from final analysis (n=42), 27% total weeks were excluded, primarily in the winter months, January – March. Examples of CyAN imagery showing pixels detecting a bloom, missing pixel data, and median pixels taken are shown in Figure 2.

The final weighted values were included in the statistical models as continuous and categorical variables. To categorize the values, we utilized the 2021 guidance by the WHO for recreational exposure to cyanobacteria, the guidance referenced by the United States and international community, which classifies an Alert Level Framework based on chlorophyll concentrations in cyanobacteria-dominated blooms (Chorus and Welker, 2021). Our weighted Chl_{BS} value was categorized according to the alert levels provided in the guidance as (1) no risk level (<3 ug/L Chl-a), (2) Vigilance Level (3–12.0 ug/L Chl-a), (3) Alert Level 1 (12.1–24.0 ug/L Chl-a), and (4) Alert Level 2 (>24.0 ug/L Chl-a). Weeks in which no pixel values were above the CI_{cyano} algorithm limit of detection were included in the lowest 'no risk' category,,and this category was used as the referent. Weeks in which no pixel values were above the CI_{cyano} algorithm limit of detection (>0.0001 units; n=27 weeks) were not included in statistical models utilizing the continuous variable.

2.4 Covariates

We considered meteorologic, temporal, and environmental variables as potential covariates. First, we considered the meteorological variables of temperature and dewpoint. We created a variable for weekly maximum temperature and weekly maximum dewpoint (in Celsius) for each study week using data downloaded from National Oceanic and Atmospheric Administration (NOAA, 2001). These factors were considered in the models as cubic polynomials for maximum temperature and continuous maximum dew point. Cubic polynomials were used to model temperature to allow for a non-linear relationship

between temperature and respiratory healthcare visits. Second, we considered time variables - year and season. Season was categorized as winter (December, January, February), spring (March, April, May), summer (June, July, August), and fall (September, October, November).

We also considered other environmental data based on a priori knowledge of their relationship with respiratory hospital visits or cyanoHAB dispersion or exposure. Air pollution variables (sulfur dioxide, ozone, and fine particulate matter [$PM_{2.5}$]) were considered because they have been previously associated with respiratory health outcomes (Eguiluz-Gracia et al., 2020; Tiotiu et al., 2020). These variables were created using data from air pollution monitors in the city of Green Bay. Daily data were aggregated into weekly averages. We did not assess wind data as a potential confounder because data for other variables in this study were aggregated by week, and we did not anticipate that wind values aggregated at the weekly level would be informative.

2.5 Statistical Methods

We used summary statistics to describe the demographics of the study population, Chl_{BS}, respiratory hospital visits, and covariates. Time series graphs showed the fluctuation of variables over the study period (2017–2019) (data not shown).

We used Poisson generalized linear models to estimate the association (rate ratio) between the weekly number of respiratory hospital visits (outcome, dependent variable) and weekly value of Chl_{BS}. Chl_{BS} was assessed as a continuous measure and a categorical measure in separate models (using the WHO Alert Levels, no alert level as the reference group compared to Vigilance, Alert Level 1, and Alert level 2). For inclusion of covariates, descriptive statistics and Spearman correlations assessed the relationship between the covariates and the health outcome (respiratory hospital visits) and primary exposure

200 (Chl_{BS}). We conducted statistical analyses using SAS v9.4 statistical software (Version 9.4 SAS Institute
201 Inc., Cary, North Carolina, USA; Power analysis was conducted using G*POWER software). Results were
202 reviewed for statistical significance using a p-value of 0.05 and considering 95% confidence intervals (CI).
203
204 2.6 Sensitivity analysis
205 We performed several sensitivity analyses. First, we added time splines to the model to consider
206 potential confounding by time-varying factors. Second, we restricted analyses to seasons with Chl_{BS}
207 present. Third, we collapsed WHO Alert level 1 and Alert level 2 due to the small number of weeks in
208 Alert Level 2, to compare to the reference category ("no risk" level) and WHO Vigilance Level. Fourth,
209 we lagged the exposure variable by one week to assess a potential lag in respiratory health outcomes
210 following Chl_{BS} presence. Finally, we examined different percentage missing cutoffs for pixels, and for
211 the continuous analysis imputed 0.001 for weeks with non-detects to determine if including those
212 weeks in the analysis changed results.
213
214 **3. Results**
215 3.1 Health data
216 We identified 2,743 hospital visits with primary diagnostic codes for asthma (n=2,448), wheezing
217 (n=140), and allergic rhinitis (n=155) during 2017–2019. Hospital visit counts per week ranged from 5 to
218 36, with an average of 17.5 hospital visits per week (standard deviation=5.76). Of the 2,743 hospital
219 visits, 56% were for female patients and 44% were for male patients. Most visits involved patients above
220 the age of 19 (65%), of White race (61%), and non-Hispanic ethnicity (86%) (Table 1).
221
222 3.2 Satellite data

223 Figure 3a shows the weekly unweighted median Chl_{BS} and the total number of pixels that detected a
224 bloom over the study period for weeks with pixels above the limit of detection (n=87 weeks). Figure 3b
225 shows the effect of weighting the Chl_{BS} by the percentage of pixels detecting a bloom, which smoothed
226 the median Chl_{BS} to account for outliers with low numbers of pixels detecting biomass. The expected
227 seasonal pattern for cyanobacteria blooms was observed, with peak values occurring between June and
228 October. When categorized by the WHO alert levels, most weeks fell within the lowest 'no risk' level
229 (n=68, 60%). There were 22 weeks (19%) in the Vigilance Level, 17 weeks (15%) in Alert Level 1, and 7
230 weeks (6%) in Alert Level 2. The categorical variable included more weeks of data because weeks with
231 no pixels above the detection limit were included in the lowest 'no risk' category (n=114 weeks).
232
233 3.3 Statistical models
234 *Covariate assessment models*
235 Weighted Chl_{BS} values are shown in Figure 4 compared to the weekly counts of hospital visits during the
236 study period. Peak of Chl_{BS} in June to October did not coincide with increased hospital visits during those
237 months. Temperature and dew point were highly correlated with Chl_{BS}, particularly when a 4-week lag
238 time was considered ($r_{S,Temperature}$=0.81, p-value <0.001; $r_{S,Dew\ point}$=0.85, p-value <0.001). Both
239 temperature and dew point were also associated with the weekly counts of hospital visits. Air pollutants
240 (sulfur dioxide, ozone, $PM_{2.5}$) were not associated with Chl_{BS} or the hospital visit outcomes and were not
241 included in the final statistical model. The final statistical model included Chl_{BS}, season, year, maximum
242 temperature, and maximum dewpoint.
243
244 *Unadjusted and Adjusted models*
245 The results of the unadjusted and adjusted statistical models assessing the association between the
246 continuous weighted Chl_{BS} and weekly respiratory hospital visits are shown in Table 2. In both the

unadjusted model and the model adjusted for season, year, maximum temperature, and maximum dewpoint, there was no statistically significant evidence of an association between Chl_{BS} and rate of respiratory hospital visits (unadjusted rate ratio (RR) [95% CI]=1.00 [0.99, 1.01]; adjusted RR [95% CI]= 0.99 [0.99, 1.00]).

Although not statistically significant, a higher rate of respiratory hospital visits was observed in weeks with Chl_{BS} values in the highest WHO alert category (Alert Level 2) than during weeks with Chl_{BS} values in the lowest category ("no risk" level) (unadjusted RR [95% CI]=1.18 [0.93, 1.48]). However, this association in the highest WHO alert category was attenuated when adjusting for covariates (RR [95% CI] = 0.98 [0.77, 1.24]) (Table 3). No associations were found in other alert levels compared to the reference levels, although the rate ratio during Alert Level 1 weeks decreased slightly, when adjusting for covariates (unadjusted RR [95% CI]=0.86 [0.72, 1.03]; adjusted RR [95% CI]=0.82 [0.67, 0.99]). We observed no differences in results from any of the sensitivity analyses completed (results not shown).

4. Discussion

This pilot study utilized existing data to explore the relationship between aerosolized cyanobacteria exposure and respiratory effects over time. This multiagency collaborative worked to develop a metric using information from CyAN satellite data to approximate cyanoHAB exposure relevant to respiratory health outcomes. Although CyAN satellite data do not provide measurements of cyanotoxins specifically, they help to contextualize the spatiotemporal distribution of cyanoHABs and the potential for cyanotoxin exposure that can be accessed consistently over time in lakes large enough to be resolvable through satellite imagery (300-m resolution). In addition, EHRs provide a cost-effective way to study specific health outcomes in large populations spatiotemporally. These two data sources allowed for the comparison of aggregated hospital visits to cyanobacteria biomass occurrence over time and could be

repeated in other populations. We also considered multiple confounding variables and validated the proposed methodology and statistical modeling through sensitivity analyses.

Although we observed no consistent associations between hospital visits and Chl_{BS}, we identified relationships that warrant further investigation. For example, our analysis showed an increased relative risk for hospital visits at the highest risk level (WHO Alert Level 2) compared to the reference category (no risk level). However, when covariates were added to the model, the relations were attenuated, and the statistical power decreased. Future work is planned to evaluate a larger geographic region with increased statistical power to better understand these relationships.

Previous studies have identified the potential for aerosolized cyanobacteria to contribute to respiratory effects. Reviews of EHR data sources that identified HAB exposure through ICD codes showed respiratory outcomes as a primary complaint for patients visiting hospitals and emergency departments (Lavery et al., 2021; Wu et al., 2021). Wu et al. (2021) examined dermal, respiratory, and gastrointestinal emergency department visits in the Greater Boston, Massachusetts whose municipal drinking water source experienced cyanoHABs during 2008–2011. Comparing emergency department visits with satellite data, Wu et al. (2021) found a positive association between cyanobacteria abundance and respiratory illness, although their study evaluated potential exposure through drinking water rather than aerosolization. Our study aimed to build upon this previous work by approximating exposure to cyanobacteria biomass through potential residential aerosolized exposure.

Our study has some limitations. First, although cyanoHABs are known to cause adverse health outcomes (Hilborn et al., 2014; Roberts et al., 2020), our study did not directly measure cyanotoxin occurrence or concentration. Instead, we used cyanobacterial recreational guidance from the WHO and a derived Chl_{BS}

295 from satellite imagery as an indicator of health risk. The Chl_{BS} algorithm, like all biomass measurements,
296 has uncertainty associated with estimates. For Chl_{BS}, uncertainty generally increases with lower
297 concentrations, which includes the WHO Vigilance Level (Seegers et al., 2021); model sensitivity to
298 algorithm uncertainty could be considered for future modeling approaches. Second, we did not account
299 for some potentially important environmental and meteorological variables such as wind direction and
300 speed and aeroallergens (e.g., pollen) in our models. Qualitatively (not included in statistical
301 calculations), weekly aggregated wind data confirmed there was little variability in wind direction during
302 our study period, nor were the winds blowing in the direction of densely populated areas most of the
303 time during peak bloom season. It is unclear whether a higher wind speed would increase the likelihood
304 of exposure by enabling aerosolized cyanobacteria or cyanotoxins to travel farther and longer, or
305 whether it would cause the bloom to separate and disperse, thus decreasing the likelihood of exposure.
306 Because the study population was south/southwest of the lake, this analysis assumed that the
307 population experience the same rate of exposure throughout the study period. Additionally, given the
308 lack of comprehensive pollen data in the study area, we did not include this data in models and
309 recognize that this may affect findings. Third, inherent limitations exist with using EHR data and using
310 ICD codes to classify health outcomes. There is substantial ambiguity in how codes are input, and in
311 some cases, combinations of codes can be used to describe a single health event, which is why we
312 included only the primary diagnosis for this analysis. In addition, due to small sample size, visits for
313 asthma, wheezing, and allergic rhinitis were combined into one count variable, which could have
314 masked associations between cyanobacteria biomass and individual health conditions. Moreover,
315 aggregating data about a weekly timepoint may have masked associations between cyanobacteria and
316 health outcomes that occur at a shorter temporal scale.

317 Future work in a larger study area that examines individual health outcomes separately would be
318 beneficial. In addition, more research using this novel exposure metric could include causal methods

particularly as it pertains to time-series analysis using a natural experimental design to better understand the observed and potential future health effects from cyanobacterial blooms. Finally, the presented analysis can be applied to other health outcomes, and possibly other environmental exposures, where existing data are available.

5. Conclusion

To protect human health and inform appropriate public health actions, it is important to better understand the significance and magnitude of health effects associated with exposure to cyanoHABs. Although there was no evidence of association between cyanoHABs and respiratory outcomes in this pilot study, we were able to use existing data and a novel metric to study cyanoHABs and respiratory health in Green Bay, Wisconsin.

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Tables and Figures

Figure 1. Study area including 2019 census population estimates for included zip codes — Green Bay, Wisconsin.

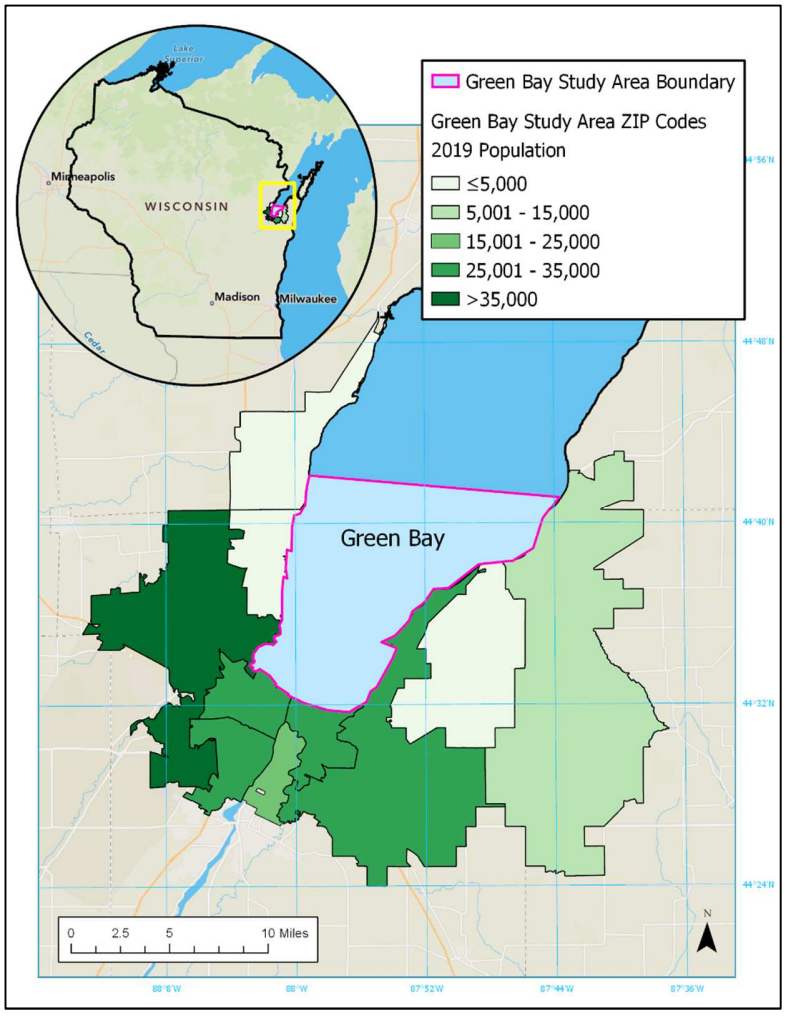


Table 1. Characteristics of patients visiting hospitals in Green Bay, Wisconsin for asthma, wheezing, or allergic rhinitis		
Characteristic	**Count**	**Percentage (%)***
Age (in years)		
<19	960	35.0
≥19	1783	65.0
Sex		
Male	1201	43.8
Female	1542	56.2
Race		
American Indian or Alaskan Native	278	10.1
Asian or Pacific Islander	41	1.5
Black	579	21.1
White	1672	61
Unknown	173	6.3
Ethnicity		
Hispanic	341	12.4
Non-Hispanic	2360	86.1
Unknown	42	1.5
*Percentage of total study population (*n*=2,743)		

Figure 2. Example CyAN images for Green Bay, Wisconsin study area boundary (pink boundary). Images show pixels detecting blooms (color), not detecting blooms (grey), and missing data (black).

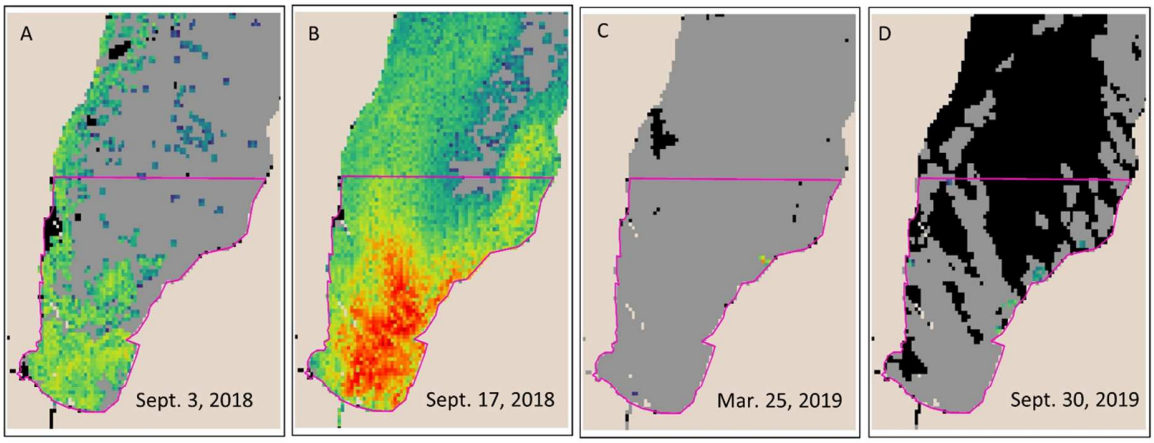


Image A shows 42% of pixels detecting a bloom and 3% with missing data. The median Chl_{BS} was 30.3 ug/L. **Image B** shows 96% of pixels detecting a bloom and 2% with missing data. The median Chl_{BS} was 58.7 ug/L. **Image C** shows 0.3% of pixels detecting a bloom and 2% with missing data. The median Chl_{BS} was 37.7 ug/L. **Image D** shows 1% of pixels detecting a bloom and 42% with missing data. The median Chl_{BS} was 7.6 ug/L.

Figure 3a. Median Chl_{BS} weekly (blue line) compared to the number of pixels detecting a bloom (dotted green line) — Green Bay, Wisconsin, 2017–2019.

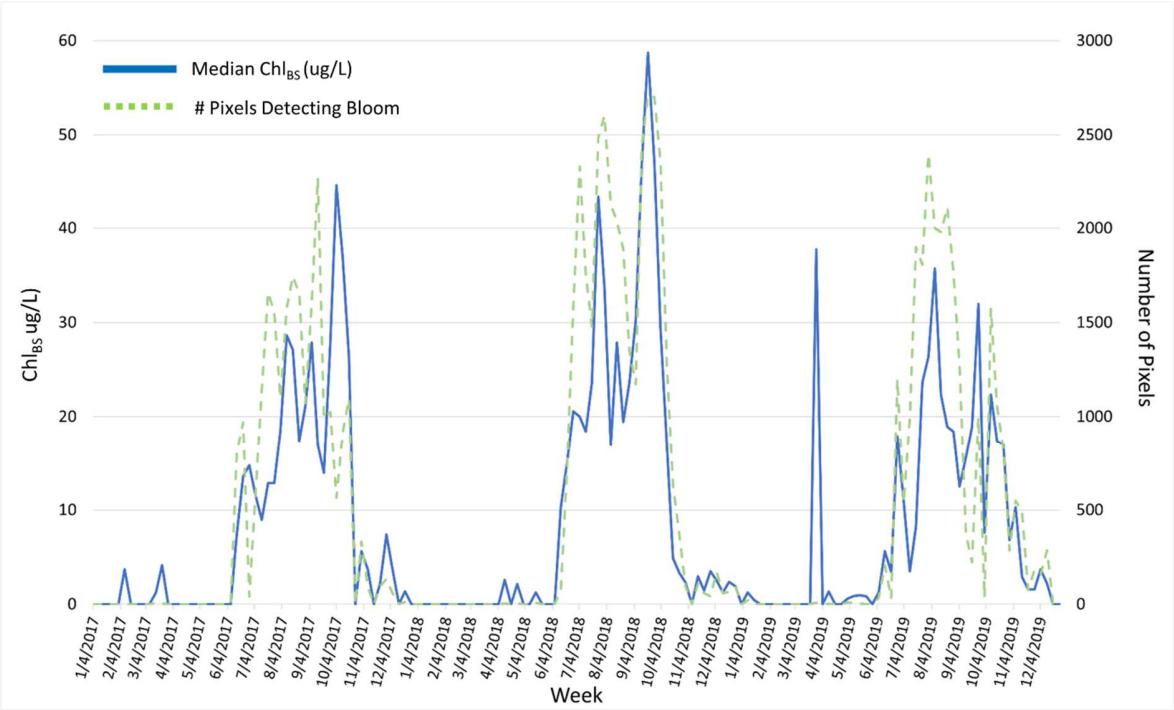
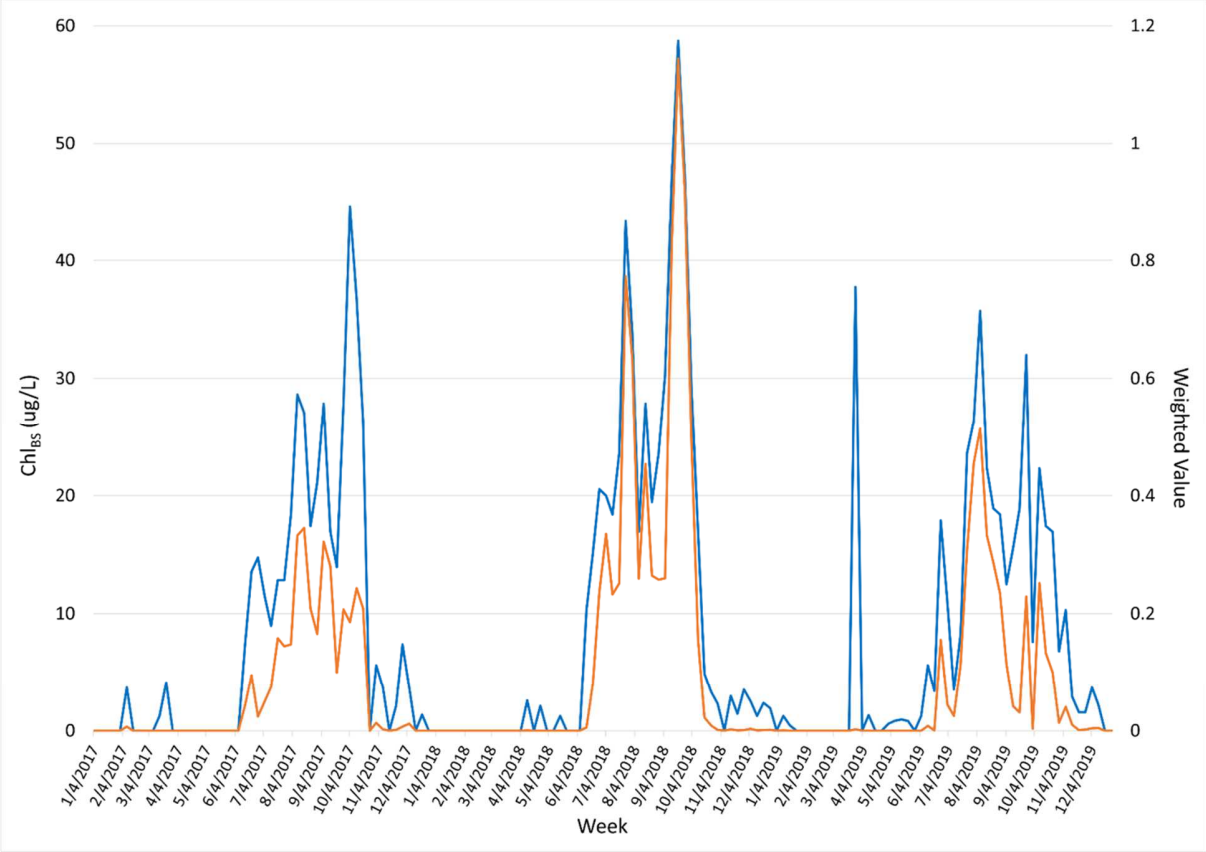


Figure 3b. Median Chl_{BS} value weekly (blue line) compared to the weighted Chl_{BS} value (orange line), Green Bay, Wisconsin, 2017—2019.



*Weighted Chl_{BS} equals the number of detection pixels divided by the total number of pixels in the study area, multiplied by the median unweighted, weekly Chl_{BS} value.

Figure 4. Weekly hospital visits and weighted median Chl_{BS}, Green Bay, Wisconsin, 2017–2019.

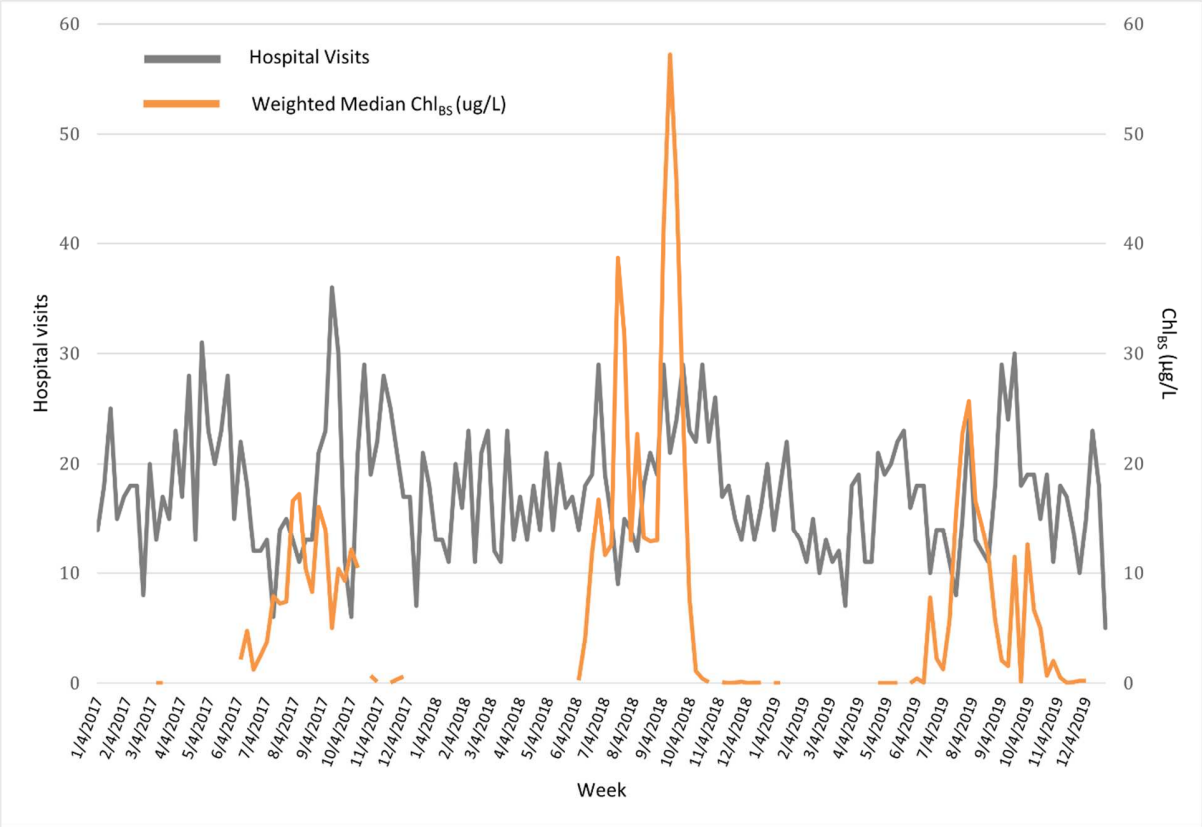


Table 2. Associations between continuous Chl_{BS} values (*n*=87 weeks), covariates, and respiratory hospital visits — Green Bay, Wisconsin, 2017–2019				
Parameter	**Rate Ratio**	**95% Confidence Interval**		P-value
Unadjusted Model				
Chlorophyll-a	1.00	0.99	1.01	0.37
Adjusted Model*				
Chlorophyll-a	0.99	0.99	1.00	0.26
Winter	0.83	0.63	1.10	0.20
Spring	0.80	0.64	1.00	0.05
Summer	0.58	0.49	0.69	<.0001
Fall (Reference)	---			
2017	0.99	0.85	1.15	0.90
2018	1.14	0.98	1.32	0.08

2019 (Reference)	---			
Temperature	1.03	0.96	1.07	0.09
Dew Point	0.99	0.95	1.02	0.43
*Adjusted for year, season, temperature, and dew point.				

Table 3. Associations between Chl_{BS} WHO categories (*n*=114 weeks) and respiratory hospital visits — Green Bay, Wisconsin, 2017–2019

WHO Category	**No. of Weeks**	**Rate Ratio**	**95% Confidence Interval**		**P-value**
Unadjusted Model					
WHO Category 1 (Reference)	68	---			
WHO Category: Vigilance Level	22	0.95	0.81	1.11	0.51
WHO Category 3: Alert Level 1	17	0.86	0.72	1.03	0.11
WHO Category 4: Alert Level 2	7	1.18	0.93	1.48	0.17
Adjusted Model*					
WHO Category 1 (Reference)	68	---			
WHO Category: Vigilance Level	22	0.88	0.74	1.05	0.15
WHO Category 3: Alert Level 1	17	0.82	0.67	0.99	0.04
WHO Category 4: Alert Level 2	7	0.98	0.77	1.24	0.86
*Adjusted for year, season, temperature, and dew point.					